

MECHANICAL DESIGN PORTFOLIO



SIVA SAI KAMMARA

MECHANICAL DESIGN ENGINEER

SOLIDWORKS

Kadapa, Andhra Pradesh, India

Phone: +91 7013029502

Email: sivasaik9999@gmail.com

LinkedIn: [linkedin.com/in/siva-sai-kammara-sk9999](https://www.linkedin.com/in/siva-sai-kammara-sk9999)



PROJECT - 03

SCREW JACK ASSEMBLY



Fig. 10 – Screw Jack Assembly

Complete 3D assembly of the manually operated screw jack developed in SolidWorks.

SOFTWARE USED

SolidWorks 2023

PROJECT TYPE

3D Part Modeling and Assembly

OBJECTIVE

Designed and developed a 3D screw jack assembly using SolidWorks by modeling individual components and assembling them to enhance CAD modeling, mechanical design, assembly, and motion simulation skills.

COMPONENTS MODELED

- Body
- Lifting Nut
- Power Screw
- Load Cup
- Washer
- Set Screw
- Tommy Bar (Handle)

WORK PERFORMED

- Created accurate 3D models of all screw jack components.
- Assembled the complete screw jack using appropriate assembly mates.
- Developed the power screw using Helix and Spiral with the Swept Cut feature.
- Applied concentric, coincident, and screw mates to achieve correct assembly motion.
- Verified smooth lifting and lowering motion.
- Generated engineering drawings.
- Developed a motion animation.

CAD FEATURES USED

- Sketch
- Extrude Boss/Base
- Extrude Cut
- Revolve Boss/Base
- Helix and Spiral
- Swept Cut
- Fillet
- Chamfer
- Circular Pattern
- Hole Wizard
- Assembly Mates
- Screw Mate
- Motion Study

SKILLS DEMONSTRATED

- 3D Part Modeling
- Mechanical Assembly
- Power Screw Design
- Motion Simulation
- Engineering Drawing
- Mechanical Design
- SolidWorks
- CAD Modeling

PROJECT GALLERY

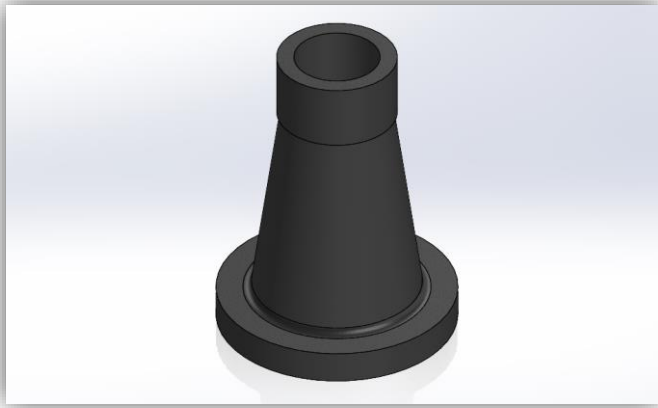


Fig. 2 – Body (Housing)

Provides structural support and houses the lifting nut and power screw assembly.



Fig. 3 – Lifting Nut

Engages with the power screw and converts rotary motion into linear lifting motion.



Fig. 4 – Power Screw

Transmits rotational motion and produces the lifting action through the threaded

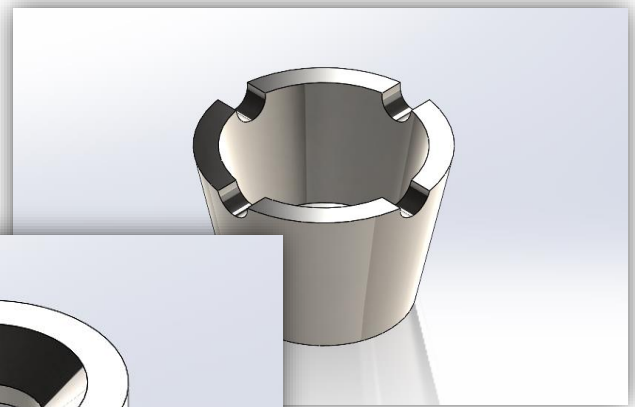


Fig. 5 – Load Cup

Supports the load and provides a stable contact surface during lifting.

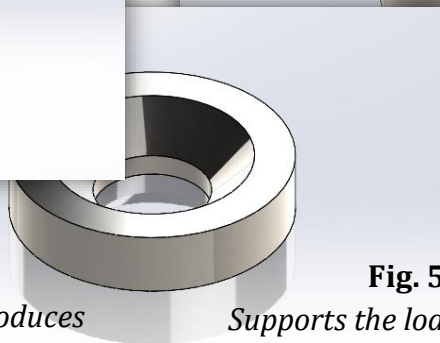


Fig. 6 – Washer

Reduces friction and distributes the applied load between mating

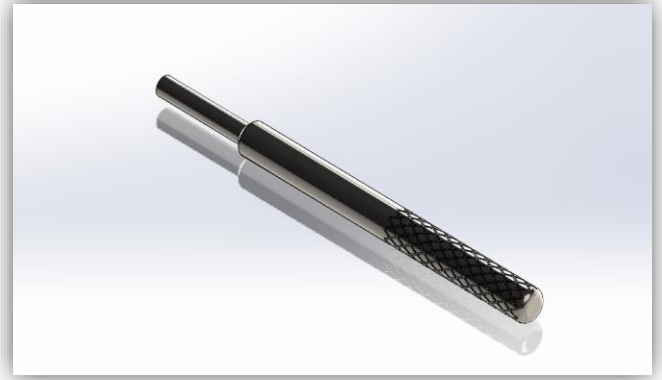
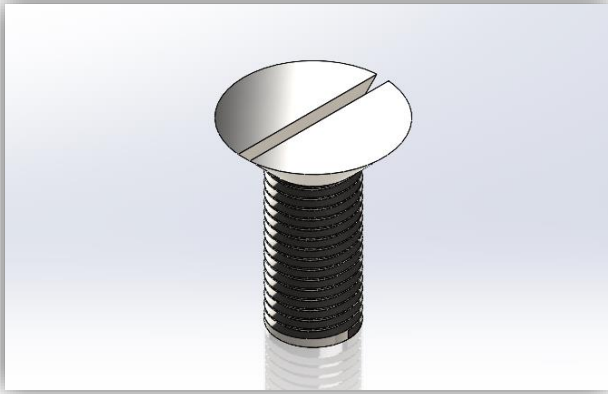


Fig. 8 - Tommy Bar

Acts as the operating handle to manually rotate the power screw.

